



INFORMATION SHEET 10

WEED AND PEST MANAGEMENT

Problems of weeds, insects and animals reduce home garden production and cause frustration for the home garden manager. Effective crop protection requires an understanding of the types of pests and diseases that attack home garden crops and an awareness of the different means of controlling them. Often, the problem will disappear if the selected plants suit their location and if the soil and water are well managed.

Plant disease is an abnormal condition caused by small organisms usually too small to be seen, called bacteria, fungi, viruses, or nematodes, which are extremely small worms that attack roots. Injury to plants is also caused by pests, such as insects, mites or worms. Some plant diseases are easily spread. For example, whiteflies transport the viruses that cause mosaic disease from diseased cassava plants to healthy ones.

Some parasitic weeds, such as *striga* and *ororbanche*, are considered pests because they cause heavy crop losses.

WEEDS

Weeds are a major problem for food plants in many home gardens. They compete with food crops for nutrients, water, sunlight and space. Weed-infested crops grow poorly and, at times, die. Densely weeded areas can harbour rats, snakes and insect pests. Weeds tend to be a bigger problem in home gardens that have no mature trees. If not well managed, weeds can require labour that could otherwise be used for cultivating useful plants.

Grass weeds can be a big problem because they can invade a garden quickly. On the other hand, they can also help a garden by preventing soil erosion and producing material for mulching and compost, or for roof thatch.

FIGURE 1

Quick-growing vine plants help reduce weed growth



How to control weeds

The main ways to control weeds are to shade them from sunlight and deprive them of a place to grow. Three techniques for doing this are to:

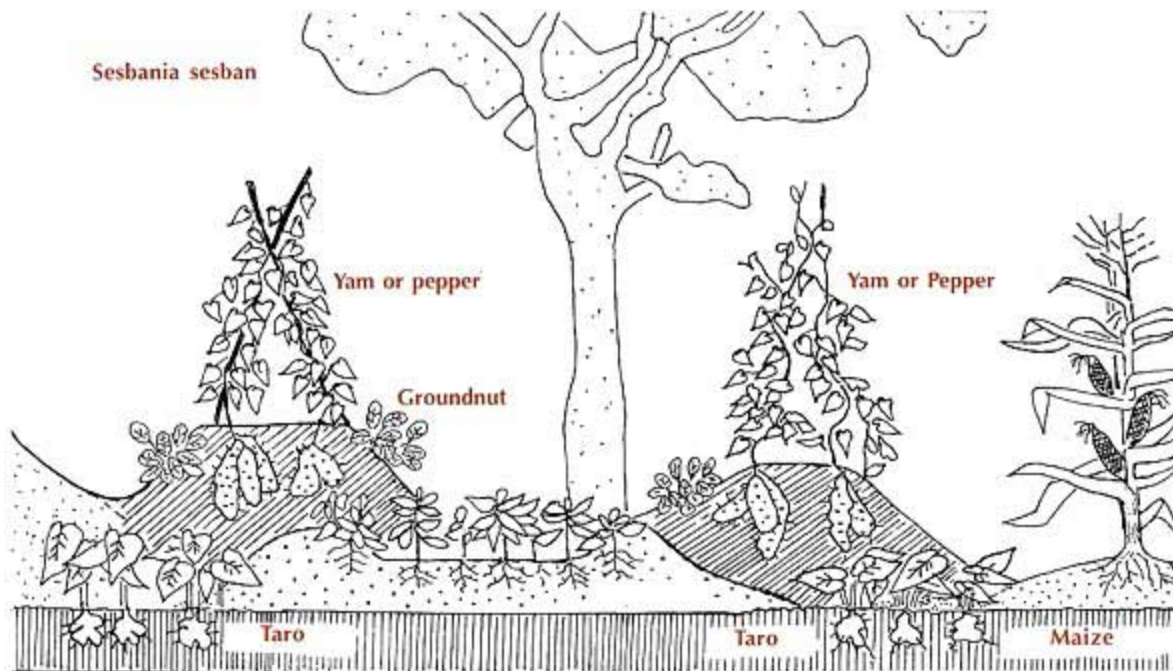
- cut or dig weeds using a hoe or sickle before the flower releases seeds;

- cover the ground with 6 cm of mulch (weeds cut by hoe or sickle can be used as mulching material);
- plant quick-growing vine plants that will cover the ground (examples of such cover crops are pumpkin, cowpea and sweet potato).

For more permanent weed control, plant crops that permanently shade the soil's surface. The multilayer cropping system, in which plants of different heights are grown together, is the most effective system.

FIGURE 2

A multilayer mixture of plants prevents weeds from receiving sunlight



PESTS AND DISEASES

Insect and disease problems tend to be seasonal, with the greatest problems occurring during the rains. In order to know when and where pest problems are likely to arise, it is important to learn the habits and life cycles of pests. Good home garden managers know as much about their plants' enemies as they know about their plants.

Insect pests. These can often be seen on plants, or in or on the soil. They damage plants mostly by chewing the leaves, roots, shoots and fruits, or by sucking sap out of the leaves, stems and fruits. Not all insects are pests. Bees facilitate the pollination of flowers. Other insects, called predators, feed on pests and reduce their population numbers.

Fungi. These attack all parts of plants. Symptoms of fungal infection are the appearance of a powdery substance on and under the leaves, rotten patches, spots on the leaves, fruit and stems, and wilting caused by affected roots. Fungi may be spread by rain or wind, or from one affected plant to another by cultivation tools.

Bacteria and viruses. These are not easily seen by the naked eye. They can cause rot in the root zones, oozing, distortion and a change of colour in the leaves and stems. Some of the symptoms are similar to those caused by fungi. Bacteria and viruses are spread through water and soil, and by affected plant materials. Unlike the rot caused by fungi, bacterial rot is usually characterized by a foul smell.

Some nematodes are soil-borne pests that attack roots, resulting in the abnormal growth of a plant's roots, stunting and generally poor growth. Affected plants often have knots and galls on their roots.

How to prevent and manage pest and disease problems

Weak plants suffer more from insects and disease than healthy plants. Good crop management, including proper attention to water, soil and weeds, will help reduce damage

from insects and disease. To prevent and manage pests and diseases in their gardens, home garden managers must adhere to these general rules.

Choose crops according to the season and the location. If an annual plant does best in full sunshine and when planted at the beginning of the dry season, the home garden manager should ensure that it is planted under these conditions. A plant grown in the wrong season or the wrong location will be more vulnerable to attack.

Select plants that grow well in the local climate. Plants from other climates may not grow well in local climates. For example, mango produces well in hot lowland areas, where the dry season coincides with the flowering and fruiting, but if it is grown in an area that is humid year round, insects and disease will damage its flowers and its fruit harvest will be poor.

Rotate vegetable crops regularly. Rotate vegetable crops, planting them a few metres away from where they grew before, and plant a different crop in the original spot (e.g. plant legumes in a place where amaranth was previously grown). This will help prevent the build-up of disease infection in the soil.

Remove diseased leaves and other parts from plants. Reducing the amount of material that insects and disease can feed on will slow their spreading.

Sprinkle wood ash. Sprinkling wood ash around the base of young plants, and also lightly on new leaf growth, deters insects and fungi.

Grow plants that are known to repel certain insects. Chilli, lemon grass, basil and marigold are common plants that produce strong smells which repel certain insects.

Use home-made pest sprays. Sprays produced from neem fruits, tobacco, chillies or soap and water deter chewing and sucking insects.

There are many synthetic pesticides that can be extremely efficient in controlling pests and diseases. However, in an African home garden, they are rarely necessary or appropriate. With the great number of crops being grown simultaneously in a home garden, the use of chemicals to control pests for one crop can easily damage another. Lack of money to buy pesticides and lack of expertise and experience in applying them - many are harmful to human health when applied incorrectly - are some of the reasons for encouraging the use of good production practices, natural control methods and available natural pesticides. When chemical pesticides must be used they should be applied judiciously and with extreme caution.

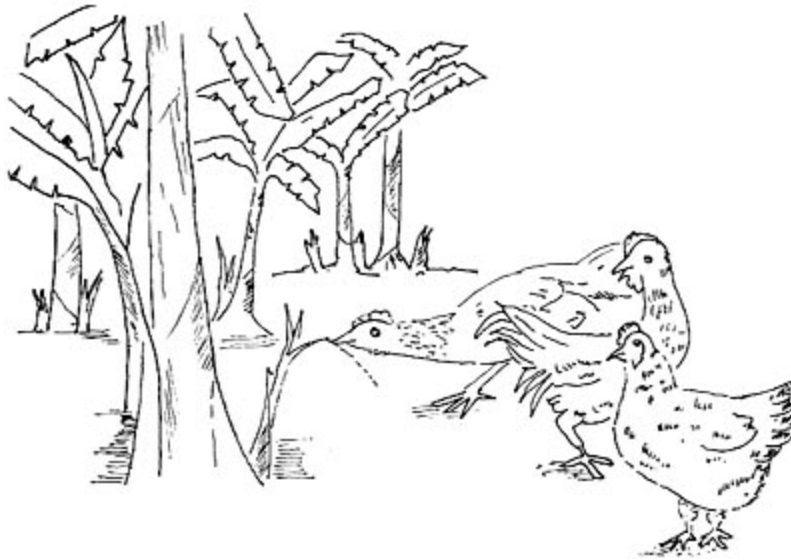
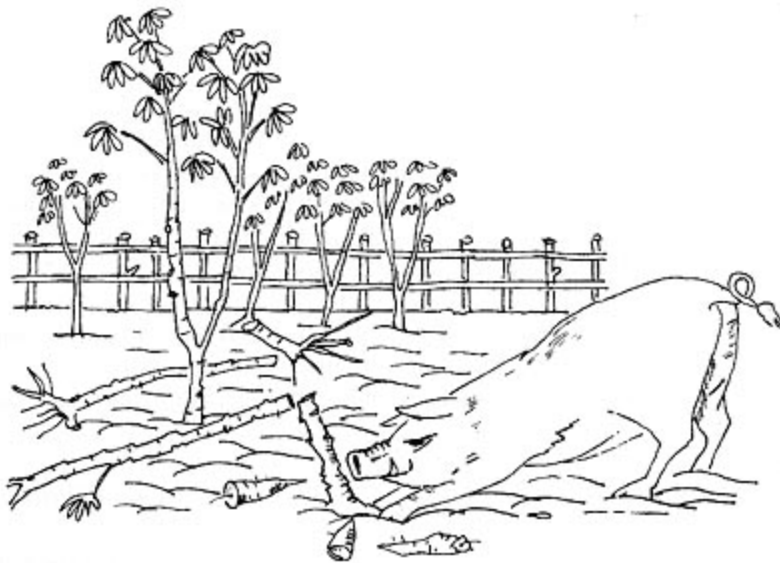
Practical information on how to control pests and diseases using natural pesticides, as well as guidelines on the safe use of chemical pesticides, if necessary, can be found in Home Garden Technology Leaflet 10, "Safe and effective crop protection".

DAMAGE FROM ANIMALS

Chickens often roam freely in home gardens. Although they catch insects and their manure fertilizes the soil, they can also become pests when they attack and seriously damage plants in their search for food. Other domestic animals, such as cattle, sheep, pigs and goats can also be a threat to home gardens. Goats and pigs sometimes roam around villages and will eat many kinds of plants. Damage from animals can discourage home garden managers to the point of abandoning home food growing.

Wild pigs, where they are not hunted for food, can be persistent and often evade farmers and cross weak fences, but *living fences* can keep them out successfully. Good methods to deter animals from entering a home garden and protecting plants from damage by animals are:

- creating a living fence of pineapple densely planted together with *Gliricidia* trees to keep out wild pigs;
- planting cassava or placing sticks around the base of a plant so that chickens cannot dig up its roots or peck at it;
- constructing a fence made from appropriate local materials;
- keeping domestic animals in fenced-in areas.

FIGURE 3**A chicken eating a banana shoot****FIGURE 4****A wild pig eating cassava**

Further weed and pest management information is included in Home Garden Technology Leaflets 9, "Cover cropping", 11, "Living fences", and 13, "Multilayer cropping".

